Public Trader Identity: Adverse Selection and Return Predictability

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Abstract

Informed traders are supposed to need anonymity: they profit by hiding among the uninformed. A decentralized exchange now publishes the counterparty. Every committed order, cancellation, rejection, and fill carries a persistent pseudonymous wallet address. We reconstruct the full-depth limit order book from a record of 17.1 billion messages and 14.3 million aggressive orders by 147,113 wallets, covering $84.3 billion in taker notional. We report three findings. First, informativeness is a persistent wallet attribute. Wallets ranked by the price movement following their aggressive orders retain that ordering across adjacent ten-day windows, with a rank correlation of 0.52. Second, the ranking predicts returns. Adding the live activity of the highest-ranked wallets to a standard anonymous benchmark of prices, quotes, and order flow raises the out-of-sample R^2 for one-second returns to 12.31%, a 13.2% gain ($t = 9.2$) that is 1.6 times the largest of 200 activity-matched placebo cohorts. Third, measured at realized trades rather than at every sampled moment, the increment grows from 1.43 to 2.47 percentage points of R^2. Public wallet histories therefore carry short-horizon price information that anonymous order-book data leave unmeasured.

Keywords: market microstructure; trader identity; limit order books; high-frequency prediction; decentralized exchanges.

1 Introduction

Anonymity is not an institutional detail in theories of informed trading; it is the condition under which the mechanism operates. The informed trader earns by being indistinguishable from the flow alongside which it trades, and because the maker cannot tell one counterparty from another, the spread charges everyone for the losses caused by a few [Bagehot, 1971, Kyle,

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1985, Glosten and Milgrom, 1985]. Decentralized exchanges now intermediate a growing share of derivatives volume, and on them that condition fails. Trading settles on a public ledger under persistent addresses, so what an account has traded is visible to everyone, permanently and at no cost. Two questions follow. First, do the same accounts stay expensive to trade against once everyone can see who they are? Second, can tracking their activity sharpen a short-horizon price forecast beyond what the anonymous book already gives?

We answer these questions on Hyperliquid, the largest decentralized exchange for perpetual futures, where the disclosure is unusually complete. It runs a conventional central limit order book with price–time priority; what is unconventional is that the book is maintained by a consensus protocol that publishes its entire input [Albers et al., 2026a]. Every committed order submission, cancellation, rejection, and fill becomes public when its block commits, carrying the persistent pseudonymous wallet address of the account that sent it. We collect this record from a non-validating node and replay the messages in consensus order, reconstructing the full-depth limit order book and linking each wallet's activity through time. Across the ten most active perpetual markets, the July 2026 record contains 17.1 billion Level-4 messages and 14.3 million aggressive orders from 147,113 wallets, representing $84.3 billion in taker notional (Table 1). The record answers both questions directly, and lets us ask a third that only arises once they are settled: where in the trading process the value of a wallet's history actually sits.

First, the same accounts do stay expensive to trade against. We score each wallet by the notional-weighted ten-second markout of its aggressive orders over ten days and freeze the ranking before evaluation; across adjacent ten-day windows the Spearman rank correlation is 0.52. The cost is concentrated in a narrow tail: after adjusting for market, time, order size, volatility, and spread, markouts remain flat through the fifteenth ventile and rise to 3.11 basis points in the top ventile. We call the wallets at the top of this ranking toxic.

Second, tracking those accounts does sharpen the forecast. On a 100-millisecond grid, we compare a prespecified ten-variable benchmark of standard predictors established in the literature, built from prices, quotes, and anonymous order flow, with the same model augmented by eleven features constructed from previously scored toxic wallets. At the one-second headline horizon, identity raises out-of-sample R^2 from 10.88% to 12.31%, a 13.2% gain ($t = 9.2$). The gain survives a broader wallet cohort, rolling and peer-adjusted scores, longer information embargoes, and a nonlinear tree model.

Two exercises ask whether the ranking itself is doing the work. The first isolates it from size and activity: we draw 200 cohorts of scored non-toxic wallets, matching each to the toxic cohort within cells formed by deciles of scoring-window notional and order count, and evaluate all 201 cohorts on the same grid, which yields 8.8 billion cohort-observations. At one second the toxic-wallet increment is 1.6 times the largest matched draw and exceeds every draw through ten seconds. The second applies the whole design, without retuning, to

an independently collected public December 2025 sample covering 26.3 billion messages and 21.9 million taker fills by 89,800 wallets; the prediction result recurs there (Appendix C).

Third, the forecasts above are evaluated at every stamp on the 100-millisecond grid, and at most of those stamps nothing trades at all. The natural question is whether identity still helps at the fills that actually occur, which are the moments that cost a liquidity provider money. The increment is larger there: 2.47 percentage points of one-second R^2 at an arrival, against 1.43 across the grid as a whole. That raises a second question, because the simplest explanation for a larger gain at arrivals is that identity is a readout of order flow: seeing historically toxic wallets resting at the quote indicates that a trade is about to arrive and in which direction, so the result would reflect speed rather than information in wallet histories. We examine that explanation in two ways. First, we reveal the realized direction of the arriving order to both models, handing the forecaster exactly what a readout of that order would supply. Second, we restrict the sample to fills after which no other parent order trades, where there is no subsequent flow to read at all. We find that neither removes the increment: revealing the direction absorbs 39% of it, and what survives that also survives the restriction. The signal is therefore not merely a reading of the arriving order's direction, nor of the trades that follow it.

Related literature. This paper speaks first to the theory and measurement of adverse selection. Because the canonical models leave the counterparty unidentified, the empirical literature infers adverse selection from anonymous data: decompositions of the spread [Glosten and Harris, 1988, Huang and Stoll, 1997], the permanent price impact of trades [Hasbrouck, 1991], the structurally estimated probability of informed trading [Easley et al., 1996], and its volume-clock successor [Easley et al., 2012]. These measures can fail exactly when identification matters most: measured adverse selection falls while genuinely informed investors trade [Collin-Dufresne and Fos, 2015]. Persistent wallet tags add an observable account-level dimension. We measure post-trade markouts by address and test their predictive content out of sample.

Second, the paper contributes to the literature on trader identity and anonymity. Studies of venues that concealed or revealed broker identifiers show that anonymity changes liquidity [Foucault et al., 2007, Meling, 2021], that order flow sorts across venues by anonymity, though not always toward the anonymous one, since on the London Stock Exchange the informed interdealer trades stayed in the direct, nonanonymous market [Reiss and Werner, 2005], and that when identifiers are visible the market learns from them [Linnainmaa and Saar, 2012]; in theory, credibly preannouncing an order can identify it as uninformed and lower its execution cost [Admati and Pfleiderer, 1991]. Confidential account-level data show that performance and information are persistent attributes of specific traders [Barber et al., 2014, Kirilenko et al., 2017, Baron et al., 2019, van Kervel and Menkveld, 2019, Hirschey, 2021], and knowing

which client is behind a trade is valuable enough that brokers leaking it lets their other clients trade against the liquidating fund [Barbon et al., 2019]. In our setting, a persistent identifier is attached to every committed order and becomes public when its block is committed, rather than remaining visible only to a broker, regulator, or researcher. We measure the identifier's value directly as an out-of-sample forecasting increment.

Third, the paper adds to the literature on the microstructure of cryptocurrency and decentralized markets, which has studied market quality and arbitrage [Makarov and Schoar, 2020, Barbon and Ranaldo, 2025], automated market makers, whose equilibrium liquidity provision has been compared directly with a limit order book [Lehar and Parlour, 2025] and whose passive liquidity loses to arbitrageurs by construction [Milionis et al., 2022], the strategic games public blockchains enable [Daian et al., 2020], and perpetual futures [He et al., 2024]. Public wallet tags have also been used to study insider activity [Félez-Viñas et al., 2022]. Recent work examines Hyperliquid's Level-4 record itself: Albers et al. [2026a] introduce the data, Albers et al. [2026b] show that rejected orders predict returns, and Barone and Lillo [2026] show that disclosed execution intentions lower trading costs, an on-venue test of Admati and Pfleiderer [1991]. A live maker experiment on the Binance perpetual documents the fill-probability/adverse-selection trade-off [Albers et al., 2025], and market-by-order clustering recovers trader types from NASDAQ order events [Zhang et al., 2025]. We shift the unit of analysis to the persistent wallet: its past markouts and its incremental forecasting content.

Finally, the results speak to market-making practice. Optimal quoting manages inventory and adverse selection [Avellaneda and Stoikov, 2008, Cartea et al., 2015], conditions on short-horizon book signals [Cartea et al., 2018]. Dealers can condition on client identity [Cartea and Sánchez-Betancourt, 2025, Cartea et al., 2026], whereas anonymous forecasting models cannot construct wallet histories from order-book data [Cont et al., 2014, Sirignano and Cont, 2019, Kolm et al., 2023]. Our forecasting comparison quantifies the predictive content lost when public wallet histories are excluded.

2 Institutional Setting and Data

Hyperliquid is the largest decentralized exchange for perpetual-futures trading. In July 2026, it averaged $6.6 billion in daily perpetual volume, accounting for 36.8% of trading across decentralized perpetual venues. The exchange runs on HyperCore, which maintains a central limit order book for each contract and matches orders using price–time priority. Order submissions and cancellations are sent to the network's validators, which collect them into short, consensus-ordered *blocks*; each block is then executed against the relevant order book. Once a block is committed, its trades, positions, and resulting book state are final. The block is therefore the atomic unit of ordering and information updates on Hyperliquid.

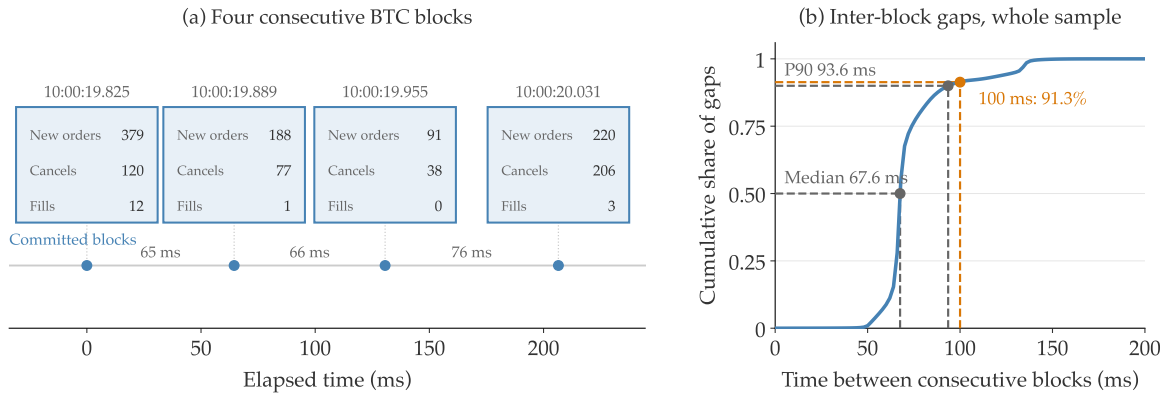


Figure 1: Committed blocks and inter-block timing. Panel (a) places four consecutive BTC blocks on July 15, 2026, at their consensus timestamps; each block reports new orders, cancellations, and fills. Panel (b) shows the ECDF of all 22,734,866 gaps between adjacent blocks in the sample.

Because blocks are committed by consensus rather than at fixed wall-clock intervals, they arrive on an irregular event clock. Figure 1 makes this clock concrete. Four consecutive BTC blocks on July 15 illustrate what each block carries and how their spacing varies (Panel (a)). Across the full sample, the median time between adjacent blocks is 67.6 milliseconds, the 90th percentile is 93.6 milliseconds, and 91.3% of gaps do not exceed 100 milliseconds (Panel (b)). At any instant, the current midpoint is therefore the close of the most recent block committed by that time. This committed-block clock underlies all prices and returns in the paper.

The distinctive feature of Hyperliquid is that all of this activity is public [Albers et al., 2026a]. Every committed block records every order attempt (submissions, cancellations, and rejections), together with every book update and execution. We collect these records continuously through a non-validating node running on a dedicated cloud server. The node connects to Hyperliquid's peer-to-peer network through bootstrap peers, synchronizes the committed chain, and writes unbuffered output locally. In parallel, we query Hyperliquid's `metaAndAssetCtxs` endpoint every ten seconds for mark prices, oracle prices, and open interest; the oracle ticks provide an external check on the reconstruction. We then replay the messages in consensus order to construct a Level-4 database of order-by-order book states, reconstructed order lifecycles, rejected order attempts, matched maker and taker executions, and the persistent pseudonymous wallet address associated with each order and trade.[1] Appendix A provides details of the collection, reconstruction, and validation procedures.

We apply this reconstruction to July 2026. Table 1 describes the ten most active perpetual contracts by message count. The sample contains 17.1 billion Level-4 messages and

[1] We use "identity" throughout for this address-level behavioral identifier. One entity may control several wallets, and one wallet may aggregate several principals; the analysis does not identify legal persons.

14.3 million aggressive orders from 147,113 wallets, representing 27.9 million taker fills and $84.3 billion in notional. Trading is led by BTC, ETH, and HYPE, with $46.8, $18.7, and $8.5 billion in taker notional, respectively. Our main analyses focus on BTC, ETH, and SOL, the three markets with the greatest message activity. As a robustness check, Appendix C replicates the main analyses using the independently collected December 2025 data of Albers et al. [2026a].

Table 1: Level-4 sample by perpetual market, July 1–27, 2026

Market	L4 messages (bn)	BBO updates (m)	Agg. orders (m)	Taker fills (m)	Wallets (thousands)	Notional ($bn)	Spread (bps)
BTC	9.75	0.94	4.02	8.66	98.9	46.8	0.20
ETH	3.03	0.91	2.04	3.89	61.7	18.7	0.62
SOL	1.35	1.69	1.35	2.16	50.5	4.9	0.26
HYPE	1.11	4.21	2.90	6.32	53.1	8.5	0.39
ZEC	0.66	6.78	1.16	2.06	26.3	2.4	0.75
LIT	0.33	4.08	1.11	1.95	14.1	0.9	2.62
WLD	0.32	4.73	0.43	0.71	18.7	0.4	2.14
XRP	0.28	0.79	0.32	0.54	21.3	0.7	1.01
NEAR	0.16	2.56	0.45	0.71	17.5	0.4	2.00
PUMP	0.09	0.58	0.51	0.89	20.9	0.5	7.13
All	17.09	27.26	14.27	27.90	147.1	84.3	0.92

Notes: L4 messages combine raw order-status and book-diff records; BBO updates are gate-valid blocks in which the best quote changed. Taker fills are individual taker legs. Because one aggressive order can execute against several resting orders, fills that share a wallet, order identifier, and market are grouped into one aggressive order. Spread is the mean quoted bid–ask spread immediately before aggressive orders. Wallets are distinct within markets and deduplicated across markets in the All row.

Markets are the ten most active Hyperliquid perpetuals by L4 message count; the All row aggregates these markets.

3 Wallet Toxicity

Every action on Hyperliquid, including each order, cancellation, and fill, carries the pseudonymous wallet address of the account behind it, allowing us to link a wallet's history to its subsequent activity. Aggressive orders are particularly informative because the trader chooses immediate execution and crosses the spread; canonical adverse-selection models therefore interpret aggressive order flow as revealing beliefs about future prices [Bagehot, 1971, Kyle, 1985, Glosten and Milgrom, 1985]. A directional trader may use aggressive orders to express a view, while a market maker may use them to hedge inventory, respond to order-flow imbalances, or act on short-term directional signals. Although these motives differ, the direction of the resulting trade may contain information about subsequent prices. This motivates our central hypothesis: wallets can be ranked by the signed markouts of their past aggressive trades, and subsequent activity by wallets with historically high markouts should help predict future returns.

This design requires persistence along two dimensions. First, wallet addresses must remain observable long enough to accumulate a useful history. Of the 2,314 wallets with at least 100 qualifying aggressive orders during July 1–10, 86.9% trade again during July 11–20. These continuing wallets account for 50.3% of all discretionary taker notional during July 1–10 and 43.6% during July 11–20 across the BTC, ETH, and SOL markets. Second, informativeness itself must persist: wallets whose aggressive trades are more informative in one period must remain more informative in the next. In the remainder of this section, we first construct our measure of wallet informativeness (Section 3.1) and then test whether it persists into a later window (Section 3.2). Section 4.2 then asks whether the activity of informative wallets adds predictive content beyond the anonymous footprint of current order flow.

3.1 Measuring toxicity

Following the market-microstructure literature, we measure post-trade adverse selection using the signed midpoint markout [Barone and Lillo, 2026, Cartea et al., 2026]. For an aggressive order e filled at time t_e with notional n_e and direction q_e ($+1$ buy, -1 sell), the ten-second markout, in basis points of the pre-event midpoint, is

$$x_e = 10^4 \, q_e \, \frac{m_{t_e+10s} - m_{t_e^-}}{m_{t_e^-}}, \tag{1}$$

where $m_{t_e^-}$ is the midpoint at the close of the last block strictly before the event's block, and m_{t_e+10s} is the latest midpoint committed by $t_e + 10s$. A positive markout means that the midpoint moves in the aggressor's direction after the fill and therefore against the passive counterparty. "Toxicity" throughout is a descriptive label for heterogeneity in this quantity; it is not a claim that a wallet corresponds to one trader or trades on private information. We use ten seconds as the benchmark horizon and report all markouts in basis points; Figure 2 also traces the markout from half a second to five minutes.

Our benchmark wallet score is the notional-weighted mean markout over a frozen ten-day window,

$$\alpha_w = \frac{\sum_{e \in w} n_e \, x_e}{\sum_{e \in w} n_e}, \tag{2}$$

computed over July 1–10 for wallets with at least 100 qualifying aggressive orders; TWAP and liquidation events are excluded. Freezing the score before model fitting and evaluation provides a transparent out-of-sample benchmark. Among 2,314 scored wallets, the top decile (231 wallets) is our primary toxic cohort; the top quintile (462 wallets) is an alternative tail cut.

The frozen window has one limitation: it gradually becomes stale and never scores wallets that first become active after July 10. We therefore also re-estimate the score each day

using the previous ten days of trading (Appendix B.3). A second alternative replaces the raw markout with a peer-adjusted markout, defined as the order's markout net of other wallets' notional-weighted mean markout in the same coin and minute (Appendix B.1). This adjustment removes price movements common to traders active at the same time. The persistence and return-prediction results remain qualitatively unchanged across the frozen, rolling, raw, and peer-adjusted definitions.

3.2 Persistence, portrait, robustness

Wallet informativeness is useful only if it persists out of sample. We use July 1–10 as the scoring window and July 11–20 as the validation window. Our persistence hypothesis is that the scoring-window ranking should continue to order wallets' validation-window markouts. Figure 2 supports this hypothesis. Among wallets that clear the 100-order threshold in both windows, scoring- and validation-window scores remain strongly aligned (rank correlation $\rho = 0.52$). Panel (a) shows that this persistence is asymmetric. The bottom decile regresses toward zero, whereas the top decile remains elevated and close to the 45-degree line. This upper-tail persistence is especially relevant because the prediction analysis tracks the top decile. Because the score is a finite-sample average, $\rho = 0.52$ understates the stability of the underlying trait. Splitting each ten-day window into odd and even calendar days and rescoring the same wallets on each half gives a split-half rank correlation of 0.52 for July 1–10 and 0.48 for July 11–20; the Spearman–Brown correction to full-window length puts the reliability ceiling at 0.68 and 0.65, so the observed across-window figure attains roughly four-fifths of what a perfectly stable trait could deliver and the implied ten-day decay in informativeness is modest.

The ventile sort provides a more direct test of whether high scoring-window ranks predict larger validation-window markouts. After controlling for market, time, order size, volatility, and spread, markouts remain flat through the fifteenth ventile before rising sharply to 3.11 basis points in the top ventile (Panel (b)). The economic content of wallet toxicity is therefore concentrated in the upper tail.

A persistent ranking could still reflect temporary price pressure generated by aggressive trades. If so, the markout should quickly reverse. Instead, the top-decile markout rises from 1.25 basis points at half a second to 2.11 basis points at ten seconds and remains elevated at five minutes (Panel (c)). The D1–D8 remainder stays near zero. Native TWAP child orders, whose timing is scheduled rather than chosen separately for each execution, provide a benchmark for mechanical execution. The D1–D8 remainder tracks this benchmark at every horizon, whereas the top-decile markout is several times larger and rises as the TWAP markout decays. This horizon profile is inconsistent with purely transient price pressure, although persistent impact, repeated same-direction flow, and latent metaorders remain possible.

Persistence is economically relevant only if the top-ranked wallets continue to trade. Table 2 shows that 91.3% of the 231 top-decile wallets trade again during the validation window. Across the fixed deciles, validation-window markouts remain sharply ordered, staying flat through D8, doubling in D9, and doubling again in D10. The result therefore reflects persistence among continuing wallets rather than attrition among the initially ranked wallets.

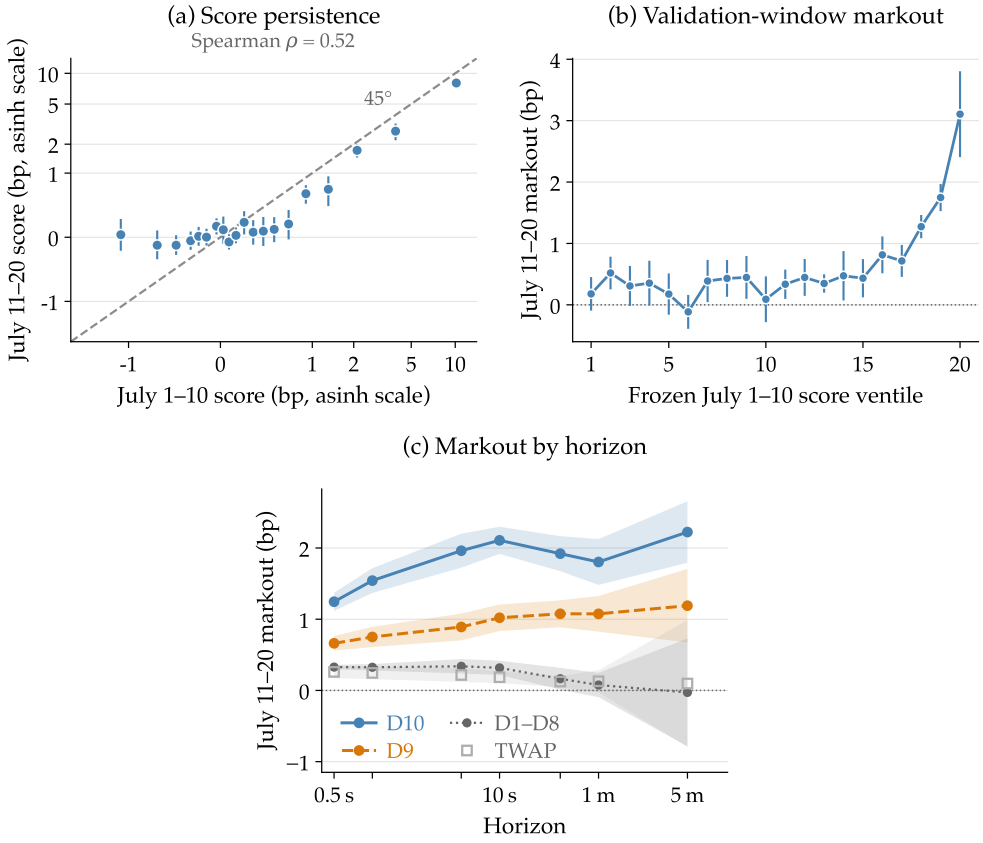


Figure 2: Persistence and horizon profile of wallet toxicity. Panel (a) compares scoring- and validation-window scores. Panel (b) reports adjusted validation markouts by scoring-window ventile. Panel (c) traces markouts for D10, D9, D1–D8, and native TWAP child orders. Bars and bands are 95% confidence intervals.

Cross-venue effects. Persistent markouts could arise mechanically if some wallets react to price changes on other venues before Hyperliquid adjusts, the cross-venue analogue of toxic arbitrage [Foucault et al., 2017]. Such a wallet should trade toward the gap between the venue's external price oracle and its local midpoint, buying when the oracle is higher and selling when it is lower. Table B.3 in Appendix B.2 shows the opposite pattern. Top-decile orders arrive on the gap's side only 41.3% of the time, compared with 57.9% for the bottom

decile, and the mean signed gap they trade into is negative. The markout also survives two controls for this gap. Adding gap controls moves the daily D10–D1 coefficient only from 1.72 to 1.75, while the subsample with a gap no larger than one basis point still yields 2.12 basis points for the top decile and 0.16 for the bottom decile. Together, these results show that cross-venue price gaps do not explain the top-decile markout.

Table 2: Wallet deciles: scoring window and out-of-sample validation

	Scoring window (July 1–10)					Validation window (July 11–20)				
Decile	Wallets	Orders	Markout (bps)	Median order ($)	Notional share (%)	Wallets	Orders	Markout (bps)	Median order ($)	Notional share (%)
D1	231	53,758	−1.13	83	0.5	203	36,380	0.27	59	0.4
D2	232	74,444	−0.39	47	0.6	195	57,109	0.06	24	0.4
D3	231	73,756	−0.20	150	0.8	203	79,110	−0.10	76	0.8
D4	232	96,900	−0.07	47	0.7	201	61,826	0.29	46	0.4
D5	231	89,907	0.06	74	0.6	193	45,794	0.03	81	0.3
D6	232	113,316	0.17	200	2.2	199	86,027	0.24	361	1.7
D7	231	72,503	0.32	982	2.5	196	40,285	0.30	790	1.7
D8	232	77,536	0.54	428	2.3	205	36,803	0.60	357	1.7
D9	231	263,740	0.96	276	11.9	204	180,469	1.09	500	11.0
D10	231	596,480	2.56	1,098	31.0	211	408,061	2.20	1,201	25.1

Notes: Deciles are formed once, on the frozen July 1–10 wallet score, and never re-sorted. Both windows report the identical five statistics of each decile's aggressive orders on the three model markets (TWAP and liquidation events excluded): distinct wallets trading, aggressive-order count, notional-weighted ten-second markout, median order notional, and the decile's share of all discretionary taker notional in that window. The share column is therefore taken against the whole window's taker flow rather than the scored subtotal, so it does not sum to one hundred percent; the residual is flow from addresses without a qualifying scoring-window history.

4 High-frequency Return Prediction with Identity

Section 3.2 shows that post-trade markouts persist across identifiable wallets. This section asks whether identifying the wallets behind market activity improves short-horizon return forecasts beyond the information available from an anonymous order book. Our design compares an anonymous benchmark with an identity-augmented model. Both observe the same public tape. The benchmark summarizes aggregate prices, quotes, and trade flow, while the augmented model adds parallel quote and flow measures for top-decile wallets, together with wallet-specific measures of trading breadth and quoted depth. The difference in predictive performance measures the incremental information revealed by wallet identity.

To make the comparison demanding, the anonymous block combines well-established predictors from several strands of the market-microstructure literature. Table 3 maps each variable to its source and groups the predictors into three families. Quote variables capture depth imbalance at the best quotes and within five basis points of the midpoint, together with quote-update order-flow imbalance over the previous one and thirty seconds [Stoikov, 2018,

Table 3: Predictor sets: the public–identity mapping

Family	Variable (code)	Reference
Panel A: paired variables (identical construction; identity restricts to the toxic decile)		
Quote	Best-quote depth imbalance (`imb` / `hotImb`)	Stoikov [2018]
Quote	Near-touch depth imbalance[c] (`nearImb` / `hotNear`)	Xu et al. [2019]
Quote	Quote-update order-flow imbalance, 1s (`ofi1s` / `hotOfi1s`)	Cont et al. [2014]
Quote	Quote-update order-flow imbalance, 30s (`ofi30s` / `hotOfi30s`)	Cont et al. [2014]
Flow	Latest signed print (`print` / `hotPrint`)	Hasbrouck [1991]
Flow	Signed taker notional, 1s (`flow1s` / `hotFlow1s`)	Biais et al. [1995]
Flow	Signed taker notional, 30s (`flow30s` / `hotFlow30s`)	Biais et al. [1995]
Panel B: public only (prices are common to all observers)		
Price	One-minute midpoint return (`ret1m`)	Andersen et al. [2003]
Price	One-minute realized volatility of 1s returns (`vol1m`)	Andersen et al. [2003]
Price	Quoted spread, bp of mid (`spread`)	Glosten and Harris [1988]
Panel C: identity only (no counterpart in the anonymous block)		
Flow	Net distinct toxic-decile buyers, 1s[a] (`hotBreadth1s`)	
Flow	Net distinct toxic-decile buyers, 30s[a] (`hotBreadth30s`)	
Quote	Toxic-decile share of best-quote depth[b] (`hotShare`)	
Quote	Toxic-decile contribution to best-quote imbalance[b] (`hotContrib`)	

Notes: Parentheses give the code identifiers used in the estimation and in Figure 4: in Panel A the anonymous variable first and its identity twin second, separated by a slash. Panel A pairs share a window, formula, and clock, the `hot` prefix restricting the construction to the toxic decile; Panel C variables exist only because wallets are distinguishable, so they carry no citation.
[a] Counts of distinct wallets; the anonymous block measures flow by signed notional rather than by trader count.
[b] Scaled by total best-quote depth: the decile's weight in the book, not its internal tilt.
[c] Depth within 5bp of the mid, in basis points of implied move (Section 4).

Xu et al., 2019, Cont et al., 2014]. Because the price move associated with a given near-touch imbalance scales with volatility [Kyle, 1985], we multiply the near-touch pair by the coin's realized one-second return volatility on the previous day and express both variables in basis points of implied price movement. Flow variables capture the latest signed print and signed taker notional over the same windows [Hasbrouck, 1991, Biais et al., 1995]. Price variables capture the previous one-minute return, one-minute realized volatility, and quoted spread [Andersen et al., 2003, Glosten and Harris, 1988].

The identity block applies the same construction to quote and flow activity from top-decile wallets. It also includes variables that can be measured only when wallets are distinguishable: the number of distinct top-decile buyers minus sellers, the decile's share of best-quote depth, and its contribution to best-quote imbalance. Price variables have no identity counterpart because prices are common to all traders.[2]

[2]The matched-wallet placebo asks whether knowing the identities of an equally active but non-toxic wallet cohort would produce the same gain. We repeat the identity construction for 200 cohorts matched to the toxic wallets on scoring-window order count and notional; none reproduces the toxic cohort's gain through ten seconds (Appendix B.4).

4.1 Design and Freezing Protocol

Return forecasting requires fixed calendar horizons, whereas Hyperliquid updates on an irregular block clock. A block-to-block target would therefore mix returns measured over different elapsed times. We instead sample the market on a regular 100-millisecond grid. This interval remains close to the venue's update frequency because 91.3% of adjacent blocks are committed within 100 milliseconds (Figure 1, Panel (b)).

Let t denote a grid timestamp. Every feature uses only information committed by t. Book and price variables are states, so they use the close of the latest block committed before t. Trade flow and quote-update order-flow imbalance are interval measures, so their values at t aggregate events in the completed interval $[t - 100 \text{ ms}, t)$. Recording the interval only at its endpoint prevents the data leakage that would arise were events after the forecast time to enter the feature vector.

For coin c and horizon Δ, the target is the forward midpoint return

$$y_{c,t}(\Delta) = \frac{m_{c,t+\Delta} - m_{c,t}}{m_{c,t}},$$

where each midpoint is the close of the latest block committed by the corresponding grid timestamp.[3] We report horizons from 200 milliseconds to thirty seconds and set targets that cross UTC-day boundaries to missing. As a robustness check to further avoid data leakage, we shift the entire feature vector back by an additional 200 and 300 milliseconds while leaving the target at t.

The protocol freezes everything in sequence: wallet scores on July 1–10; model fitting on July 11–20 (ridge with coin intercepts as the benchmark; features standardized with fitting-window moments; penalties by expanding-day validation); one evaluation pass on July 21–27.[4] We report $R^2 = 1 - \sum_{c,t}(y_{c,t} - \widehat{y}_{c,t})^2 / \sum_{c,t} y_{c,t}^2$ against a zero forecast, equal-weighted per grid cell, with inference from the seven paired daily out-of-sample MSE differences.

4.2 The Incremental Value of Identity

If wallet histories contain information absent from the anonymous tape, adding identity should improve forecasts on dates untouched by model selection. Consistent with this hy-

[3] The same block-close rule governs Equation (1). Its pre-event midpoint uses the last block strictly before the event's block and must be at most 60 seconds old. The endpoint uses the last block committed by $t_e + h$ and is missing if that state is more than 120 seconds old, outside the sample, or inside a gate-suppressed interval. The prediction grid carries the latest emitted state forward and does not apply these markout-specific staleness screens. All timing is indexed by consensus timestamps on the committed tape rather than by a separately audited local receipt-time clock.

[4] The window ends July 27 because the collection infrastructure, not the market, sets the boundary: the collector fell behind the chain during July 27 and the node failed outright at 06:06:45 UTC on July 28. This documented infrastructure event was fixed independently of any model output and is detailed in Appendix A.1.

Table 4: Return prediction, top-decile identity block

Horizon	Ridge (benchmark)				Gradient-boosted trees			
	Anonymous R^2 (%)	+ Identity R^2 (%)	Gain (%)	t	Anonymous R^2 (%)	+ Identity R^2 (%)	Gain (%)	t
0.2s	6.91	8.20	+18.7	8.1	19.26	21.32	+10.7	5.4
0.5s	10.16	11.76	+15.8	8.3	21.00	22.94	+9.2	4.6
1s	10.88	12.31	+13.2	9.2	19.48	20.65	+6.0	5.0
2s	10.60	11.66	+10.0	8.5	16.35	17.04	+4.2	6.0
5s	9.17	9.79	+6.7	9.5	12.00	12.36	+3.0	6.1
10s	7.18	7.56	+5.2	9.9	8.55	8.76	+2.5	6.4
30s	3.56	3.67	+3.0	6.1	3.74	3.84	+2.7	1.6

Notes: Frozen wallet score, top decile—the paper's headline specification. Forward midpoint returns on the 100ms grid, three model markets; scores July 1–10, models fit July 11–20, evaluation July 21–27, with trees trained July 11–19 and July 20 held out for early stopping.
Anonymous is the benchmark without identity features and *+ Identity* adds the identity block; both learners use the same variable blocks. The *Anonymous* and *+ Identity* columns report out-of-sample R^2, Gain $= R^2_{+\text{Identity}}/R^2_{\text{Anonymous}} - 1$ is the relative increase, and t comes from the seven paired daily MSE differences. Horizons past thirty seconds are omitted, the anonymous R^2 falling toward two per cent there.

pothesis, the identity-only model produces an out-of-sample R^2 of 10.06% at one second under the frozen score, compared with 10.88% for the anonymous block, even though the identity features observe only the toxic decile's activity. Combining the blocks raises R^2 to 12.31%, a 13.2% relative gain ($t = 9.2$). Table 4 shows that this improvement is not confined to the headline horizon. The gain remains distinguishable from zero through thirty seconds. We limit the reported horizon to thirty seconds because the anonymous R^2 subsequently falls toward 2%, making relative gains on the small denominator difficult to compare with the short-horizon results.

Gradient-boosted trees extract substantially more information from the anonymous features, raising their one-second R^2 to 19.48%. Even against this stronger nonlinear benchmark, identity raises R^2 to 20.65%, a 6.0% relative gain ($t = 5.0$). The smaller gain than under ridge indicates that nonlinearities in the anonymous features capture part, but not all, of the information supplied by identity. The gain under trees declines from 10.7% at 200 milliseconds to 2.7% at thirty seconds. The identity gain remains distinguishable from zero through ten seconds, but not at thirty seconds ($t = 1.6$).

The result is also insensitive to how wallet scores and the toxic cohort are defined. Re-estimating scores on a rolling ten-day window produces a 12.7% one-second gain, close to the 13.2% gain under the frozen score (Table B.4). Expanding the cohort from the top decile to the top quintile leaves the one-second gain essentially unchanged, indicating that the predictive content is concentrated in the top decile.

Order-flow toxicity should matter most immediately after information enters the market and fade as the market absorbs it. The horizon profile in Figure 3 supports this view. Overall

forecast fit is mildly hump-shaped, rising over the first few hundred milliseconds, peaking between 0.5 and one second, and declining thereafter. This short effective horizon is consistent with Kolm et al. [2023], who find that stock-specific order-book forecasts remain effective for approximately two average price changes. The incremental value of identity decays more steadily: its relative gain falls from 18.7% to 3.0% under ridge and from 10.7% to 2.7% under trees between 200 milliseconds and thirty seconds. Identity therefore captures fast-moving information rather than a persistent wallet characteristic that happens to correlate with returns.

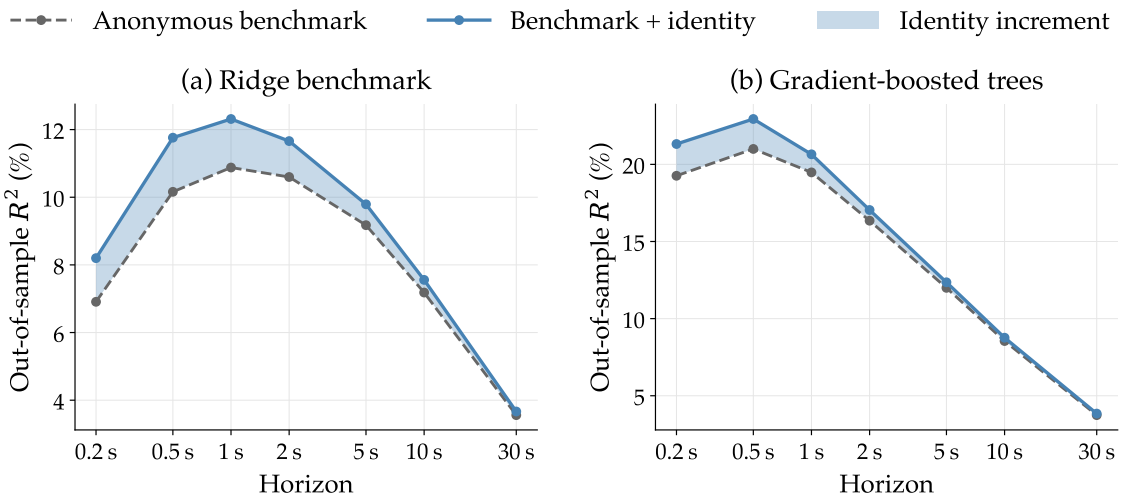


Figure 3: Out-of-sample R^2 by horizon under ridge and gradient-boosted trees. Gray is the anonymous benchmark, blue adds the top-decile identity block, and shading is the identity increment.

Where the identity value lives. The aggregate gain could arise from a new identity-specific signal or from identity variables substituting for anonymous predictors. Figure 4 distinguishes these possibilities by removing each variable in turn, refitting the one-second ridge model, and recording the loss in out-of-sample R^2, in percentage points. Because the regressors are correlated, these contributions are descriptive and non-additive. In the anonymous benchmark, the largest contribution comes from near-touch depth imbalance (`nearImb`, 1.93); once identity is added its contribution falls to 0.24. The identity-side contributions are concentrated in the toxic decile's one-second quote flow (`hotOfi1s`, 0.67) and net buyer breadth (`hotBreadth1s`, 0.37). This pattern is consistent with overlap between anonymous near-touch imbalance and the cohort-specific quote variables, not a causal decomposition.

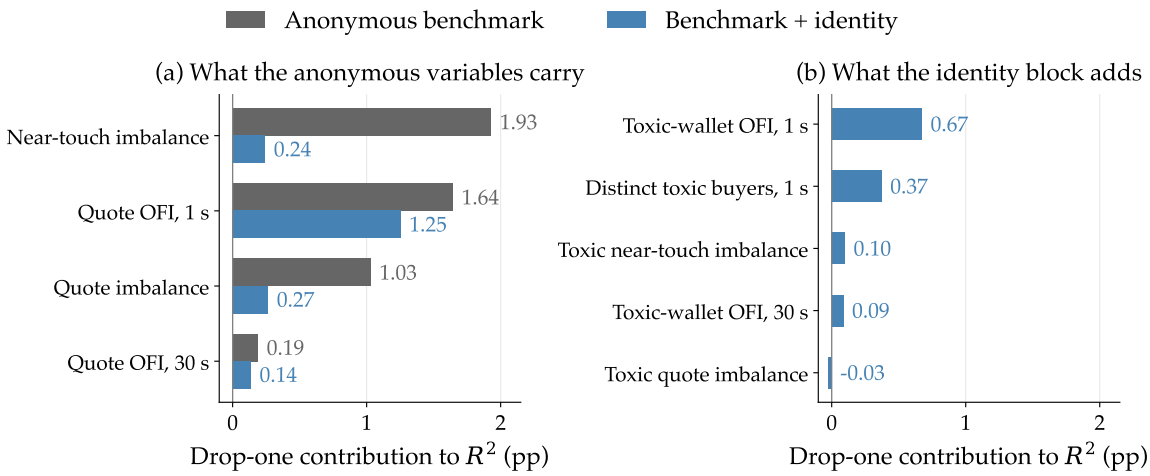


Figure 4: Drop-one contributions at the one-second ridge benchmark. Panel (a) compares anonymous predictors before and after adding identity; Panel (b) shows identity predictors. Contributions are descriptive and non-additive; each anonymous variable is shown with its identity counterpart, and pairs in which neither member reaches an R^2 contribution of 0.05 percentage points are omitted.

Matched-wallet placebo. Toxic wallets are unusually large and active, so identity features built from any comparably active cohort might improve forecasts. To isolate the toxicity ranking from size and activity, we construct 200 cohorts from scored non-toxic wallets, matching each draw to the toxic cohort on deciles of scoring-window notional crossed with order count. We then rebuild all eleven identity variables for every cohort from the block-level book and flow records and carry them to the same 100-millisecond grid. In total, the exercise evaluates the real and 200 matched cohorts at 44.0 million grid stamps across fifty-one coin-days, yielding 8.8 billion cohort-observations from 126 gigabytes of replayed per-wallet quote tapes.

At the one-second headline horizon, the toxic-decile identity increment is 1.6 times the maximum among the 200 matched cohorts. The toxic cohort exceeds every draw at every horizon through ten seconds (rank $p = 1/201$). Figure 5 shows this separation for both tail cuts, and Table B.5 in Appendix B.4 reports the exact increments and ranks. At thirty seconds, the top decile enters the placebo range, while the top quintile continues to exceed every draw. The toxicity ranking therefore adds information beyond size and activity precisely at the horizons where identity is predictive.

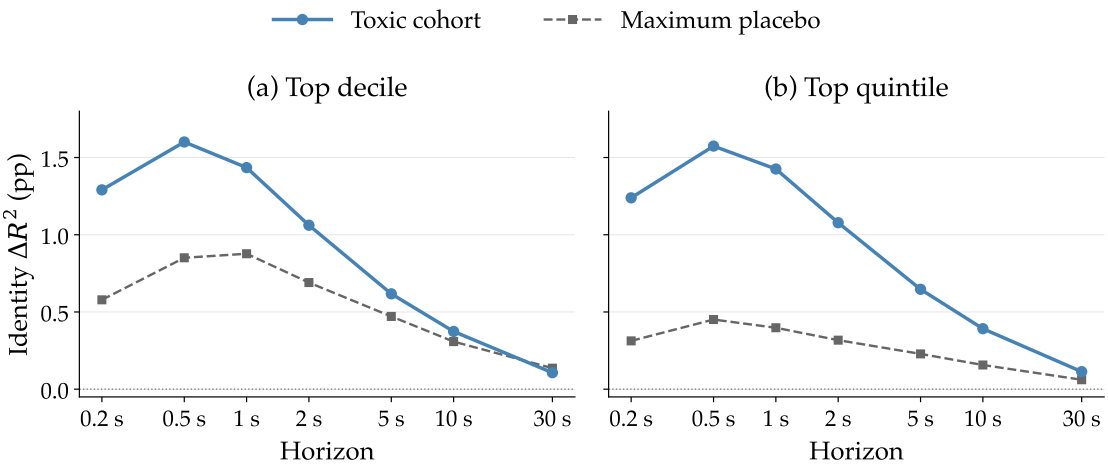


Figure 5: Identity increment for the toxic cohort against the maximum across 200 activity-matched placebo cohorts, by horizon.

Feature embargo. A second concern is mechanical rather than economic: the grid samples the last committed block, so an increment could in principle reflect a hundred milliseconds of misalignment rather than information. We test this directly by delaying *every* feature, including both the anonymous and identity blocks, by 200 and by 300 milliseconds while leaving the target at the grid stamp, so that the endpoint of a one-second forecast lies 1.2 or 1.3 seconds after the newest information. If misalignment explains the identity gain, these delays should make it collapse. Instead, Table 5 shows a smooth decay under both learners: under ridge the one-second gain falls from 13.2% to 10.2% to 9.1%, with $t = 8.2$ under both delays, and under trees from 6.0% to 4.5% to 4.0%. This smooth decay is inconsistent with a mechanical misalignment explanation. The delay reduces the anonymous benchmark far more than the identity increment. At one second, the benchmark falls from 10.88% to 6.40% under ridge and from 19.48% to 9.64% under trees. This disproportionate decline is consistent with the public order-book state becoming stale faster than the identity increment.

4.3 Identity and Subsequent Order Flow

The forecasts in Section 4.2 are evaluated at every stamp on the 100-millisecond grid, and at most of those stamps nothing trades. Whether identity still helps at the fills that actually occur is a separate question, and it is the one that matters for a liquidity provider, since those are the moments at which a resting quote is taken. The increment is larger there: 2.47 percentage points of one-second R^2 at an arrival, against 1.43 across the grid as a whole. Forecasting is also easier at a fill, where the anonymous benchmark is about twice as strong, so the comparison is one of explained return variance rather than of relative gain.

Table 5: Feature embargo: identity gain when every feature is stale

Horizon	No embargo			Features delayed 200 ms			Features delayed 300 ms		
	Anonymous R^2 (%)	+ Identity Gain (%)	t	Anonymous R^2 (%)	+ Identity Gain (%)	t	Anonymous R^2 (%)	+ Identity Gain (%)	t
Panel A: Ridge (benchmark)									
0.2s	6.91	+18.7	8.1	4.74	+14.6	8.3	3.49	+12.2	8.1
0.5s	10.16	+15.8	8.3	6.51	+12.4	9.1	5.23	+10.8	9.1
1s	10.88	+13.2	9.2	7.61	+10.2	8.2	6.40	+9.1	8.2
2s	10.60	+10.0	8.5	8.02	+8.0	8.8	7.04	+7.2	8.5
5s	9.17	+6.7	9.5	7.55	+5.5	9.0	6.89	+5.1	7.5
10s	7.18	+5.2	9.9	6.14	+4.4	9.4	5.71	+4.2	9.1
30s	3.56	+3.0	6.1	3.14	+2.6	5.4	2.96	+2.4	5.3
Panel B: Gradient-boosted trees									
0.2s	19.26	+10.7	5.4	8.85	+10.3	4.6	6.00	+6.2	4.8
0.5s	21.00	+9.2	4.6	11.14	+6.1	4.8	8.42	+5.0	5.2
1s	19.48	+6.0	5.0	11.97	+4.5	5.2	9.64	+4.0	6.0
2s	16.35	+4.2	6.0	11.40	+3.5	5.8	9.69	+3.3	5.6
5s	12.00	+3.0	6.1	9.44	+2.7	6.1	8.48	+2.3	4.1
10s	8.55	+2.5	6.4	7.10	+2.4	5.6	6.52	+2.3	4.2
30s	3.74	+2.7	1.6	3.26	+3.8	3.5	3.07	+3.1	1.6

Notes: The embargo delays every feature—the anonymous block and the identity block alike—by 200 ms or 300 ms, while the target is left at the grid stamp, so the endpoint of a one-second forecast lies 1.2 or 1.3 seconds after the newest information. Frozen wallet score, top decile; sample, variable blocks, learners and t construction are those of Table 4. *Anonymous* is the benchmark without identity features; Gain $= R^2_{+\text{Identity}}/R^2_{\text{Anonymous}} - 1$ is the relative increase over that learner's own benchmark, and t comes from the seven paired daily MSE differences.

The simplest explanation of that larger increment is that it is a readout of order flow: seeing historically toxic wallets resting at the quote indicates that a trade is about to arrive and in which direction, and price follows the trade. Under this interpretation, the result reflects speed rather than information in wallet histories, and the increment should disappear once order flow is accounted for. We examine this explanation in two complementary ways. The first reveals the direction of the focal arriving order; the second restricts the sample to fills followed by no trade from another parent order.

Both exercises begin with the 3.09 million maker fills in the three model markets over July 21–27, each matched to the last grid observation at least 100 milliseconds before its consensus fill time, so that every feature precedes the fill on the committed-tape clock. Models are fitted on July 11–20 fills, and the arriving taker's wallet address is never a predictor.

Revealing the direction. The first exercise adds the realized side $q = \pm 1$ of the focal order to both models and interacts it with every predictor, allowing predictor slopes to differ between buyer- and seller-initiated fills. Because side is realized after the feature timestamp, this is an ex post decomposition rather than a feasible forecast. At one second, identity raises R^2 by 2.47 percentage points before side is observed ($t = 10.8$) and by 1.50 points once side

is revealed ($t = 8.1$), a reduction of 39%. The conditional increment becomes statistically indistinguishable from zero at ten seconds (Table 6).

Table 6: Identity increment after revealing side and restricting subsequent trades

	Pre-arrival		Realized side revealed		No other parent order		
Horizon	ΔR^2 (pp)	t	ΔR^2 (pp)	t	ΔR^2 (pp)	t	Fills (000)
0.5s	+2.53	7.4	+1.49	4.0	+1.09	7.9	1,492
1s	+2.47	10.8	+1.50	8.1	+1.04	5.5	817
2s	+1.79	4.9	+1.41	10.6	+1.03	2.5	347
5s	+1.47	3.5	+1.26	5.9	+0.49	0.9	86
10s	+0.87	2.1	+0.63	1.6	+0.04	0.2	20
30s	+0.45	1.8	+0.35	1.4	+0.04	0.5	2

Notes: Identity ΔR^2 is the change in out-of-sample R^2 from adding identity features to the anonymous benchmark for the forward midpoint return at the stated horizon. Each of the 3.09 million evaluation-window maker fills is paired with the last nominal 100ms grid timestamp at least 100ms before its consensus fill time, and inputs use the last commit available by that timestamp.

Pre-arrival uses only predictors observed before the focal fill and excludes its side. *Realized side revealed* adds $q = \pm 1$ to both models and interacts it with every predictor; this is an ex post decomposition. *No other parent order* restricts both fit and evaluation to fills followed by no trade from a different parent order within the horizon. Further fills from the focal parent order are allowed, and this column does not condition on q. The subsample is defined by a realized future property and shrinks sharply with the horizon, which is why its fill count is reported. All t statistics come from seven paired daily differences.

Realized side therefore accounts for a meaningful share of the identity increment but does not summarize it. Over the second beginning at the fill, and counting the arriving order itself, the sign of net signed taker flow agrees with q for 86.6% of fills, while q explains only 4.0% of that flow's variance. The exercise provides a strong control for flow direction but leaves variation in order size, sweep intensity, and subsequent flow.

Restricting subsequent flow. The side of the focal order need not summarize the orders that follow it. After excluding the focal parent order, the sign of residual one-second flow agrees with q only 57.8% of the time. We therefore repeat the prediction comparison among fills followed by no trade from a different parent order within the forecast horizon. This restriction holds for 26.4% of evaluation fills at one second and 48.3% at half a second. Further fills from the same parent order remain part of the focal event.

Within this restricted sample, identity raises R^2 by 1.04 percentage points at one second ($t = 5.5$, 816,944 observations) and by 1.09 points at half a second ($t = 7.9$, 1,491,847 observations). Because the restriction uses a realized future outcome, it is a mechanism diagnostic rather than a feasible forecast. It also changes the market composition and becomes increasingly selective at longer horizons, leaving 86,000 fills at five seconds and 20,000 at ten

seconds.

Together, the two exercises narrow an order-flow explanation without ruling it out. Identity's predictive content is not summarized by the focal order's direction, and it remains among fills followed by no other parent order. The tests do not distinguish anticipation of the focal order's size and impact from subsequent quote revision. They show that wallet histories contain information about the price response that is not captured by the anonymous market state or the focal order's side alone. Translating this forecasting increment into implementable profits would require a dynamic execution and market-making model accounting for latency, queue priority, fill probabilities, fees, and inventory, which is beyond the scope of this paper; Appendix D instead reports the increment in payoff units.

5 Conclusion

Hyperliquid's public Level-4 record makes it possible to reconstruct the limit order book while linking trading activity through persistent wallet addresses. Using this structure, we document substantial persistence in address-level markouts: the rank correlation of wallet toxicity across adjacent ten-day windows is $\rho = 0.52$. This persistence also translates into return predictability. In the three most active markets, adding features constructed from previously scored toxic wallets raises one-second out-of-sample R^2 by 13.2% relative to the prespecified anonymous benchmark in a linear ridge model. Both findings recur in an independent December 2025 sample.

The analysis focuses on aggressive fills and quotes, but the same transparency opens several directions for future research. Wallet-level cancellations and rejected orders may contain information not captured by executed trades and resting quotes. More broadly, linking the full sequence of maker and taker decisions provides a foundation for evaluating address-based trading strategies in a dynamic market environment with latency, order choice, queue priority, fill risk, fees, and inventory constraints.

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A Details on the Data

This appendix documents how the Level-4 record was collected and reconstructed, the invariants the reconstruction enforces, and the diagnostics that make both auditable. The guiding distinction is between *coverage* (what was collected, established by block counts, gaps, and archive hashes) and *reconstruction validity* (whether an emitted book state agrees with an independent contemporaneous oracle). Fill pairing and replay diagnostics provide additional checks. We report each separately.

A.1 Collection

A non-validating Hyperliquid node (`hl-node`, version pinned in the replication artifact; managed by the exchange's visor process) runs continuously on a dedicated cloud instance (8 vCPUs, 64 GB memory), connects to the peer-to-peer network through bootstrap peers, and writes five raw streams as they arrive: order statuses, raw book diffs, fills, TWAP statuses, and miscellaneous events. Each stream is stored as hourly zstd-compressed NDJSON whose lines are *block envelopes* with fields `local_time`, `block_time`, `block_number`, and `events`—so the consensus block is the atomic unit of the archive, and record-level data are the exploded `events` arrays. In parallel, the `metaAndAssetCtxs` API is sampled every ten seconds for mark, oracle, and open-interest state. The July 1–30, 2026 collection window contains two material infrastructure events. On July 27 the collector fell behind the chain from 13:30 UTC, peaking at thirty-five minutes of lag, and while that far behind it shed blocks outright, leaving two holes in the tape of thirty-three and thirty-one minutes beginning at 15:05:23 and 17:05:34 UTC; it had recovered by 22:10 UTC. The oracle gate closed for all of the first hole in all three markets and for the second's final seventeen minutes, so that time is suppressed rather than emitted. During the opening fifteen minutes of the second hole the reconstruction held a stale book that still agreed with the oracle to within tolerance—the staleness limitation of a midpoint gate rather than a reconstruction error, and the reason we report the gate as validating the midpoint and not the book. The collector then ran normally until it failed outright at 06:06:45 UTC on July 28, and service resumed at 04:23:20 UTC on July 29; from July 30 a second, independently operated node provides the tape, spliced at 20:00 UTC (primary node through hour 19, second node from hour 20). All analyses in the paper end on July 27; the July 29–30 extension, where used, is stated explicitly.

Media damage and recovery. Fourteen archived source files suffered disk CRC read errors: eleven order-status hours on July 3–7 and July 21, and three book-diff hours on July 20. GNU `ddrescue` recovered them non-destructively: 18.36 GB of readable bytes were retained and 16.78 MB remain unresolved across 1,337 mapped gap ranges, zero-filled in the recovered copies and never assumed byte-complete. Nine book-diff files with verified deterministic decompression failures are consumed only up to their probed last good block; their unreadable

suffixes are the only lost data, and each creates a replay seam. Together with the collection and node boundaries, these seams produce thirteen reconstruction segments. Downstream, the reconstruction treats damage conservatively: any unexpected decompression error aborts the segment (the whitelist covers exactly the nine probed files), the book restarts *empty* at the first intact post-gap file, and nothing is emitted until a contemporaneous-oracle gate readmits the state—twelve consecutive ten-second oracle ticks agreeing with the reconstructed mid within 25 basis points, with any 30-second tick gap closing the gate and resetting the streak. Later gate failures suppress new state until the gate reopens. This is gate-revalidated BBO reconstruction, not exact full-book recovery: latent resting-order errors can survive while the mid agrees, a limitation we disclose rather than assume away.

A.2 Reconstruction conventions and checks

Reconstruction proceeds in five deterministic steps, each chosen so that the output is a pure function of the consensus-ordered block sequence.

(i) Segmented replay under a zero-corruption policy. The book is rebuilt by replaying order-status and book-diff events block by block. Each segment starts from an empty book; no state is emitted until the oracle gate admits the reconstruction. An unexpected decompression or parse error aborts the segment rather than skipping records; each of the nine verified damaged files is consumed only through its last good block. *(ii) Block-close state.* A midpoint or depth quantity evaluated at any instant t is the block-close state of the last block committed at or before t—a right-continuous, backward-looking step function. *(iii) Fill pairing.* Every execution produces one record per side; pairing on (day, market, trade identifier) recovers maker and taker for each fill. *(iv) Aggressive-order grouping.* Fills sharing (wallet, order identifier, market) are one aggressive order—one trading decision. *(v) Lifecycle linkage.* Order-status events link placement, modification, cancellation, and execution by order identifier within the collected stream.

Replay is applied in completed-block order. A block-number rewind or crossed book triggers a book reset and closes the gate; aggregate depth is kept non-negative. Reconstructed state is emitted only while its midpoint agrees with the independent oracle. This gate validates the midpoint, not every latent resting order or the full depth profile, so full-book errors can survive while the midpoint agrees.

Two identity variables are shares of book depth—the toxic decile's weight at the touch and its signed contribution to the touch imbalance—and are therefore bounded by one in absolute value by construction. On a handful of coin-days the top of book prints momentarily near-empty, and a ratio computed against that denominator can exceed its own bound. We clamp the denominator to be at least the numerator, which enforces the bound at every observation, and the feature builder asserts it before writing. Five coin-days inside the sample are affected,

and all reported results use the corrected features.

Gate closures are rare, and where they occur they are shared across markets. Over July 1–27 the gate is closed for 0.97% of coin-time in the three sample markets, rising to 1.83% inside the July 21–27 evaluation window. Across the full sample the median closure lasts 123 seconds—the twelve-tick re-admission streak itself, so the typical closure is a brief re-warmup after one disagreeing tick—while suppressed time is dominated by a short tail of long episodes, the longest running 70 minutes. Those episodes are market-wide rather than coin-specific: 81% of suppressed time in the three sample markets is common to all three at once, which is what an oracle-side outage looks like and not what selective reconstruction failure would look like. The distinction matters, because state that went missing precisely when one book was hard to rebuild would bias the sample, whereas losing the same clock window in every market does not. The replay also logs its own discontinuities—eight node-restart rewinds across the production segments, each forcing a book reset and gate re-admission, and 132 crossed-book self-heals—and a second reconstruction of the same raw stream, run separately, reproduces both counts segment by segment.

That the segment boundaries themselves are harmless can be checked directly. July 12 lies deep inside a continuous six-day segment, so its book was built by a replay that had been running for two days. Replaying July 12 on its own—empty book, warm-up, gate re-admission—reproduces the continuous replay's bid, ask, mid, and spread on 99.994% of the 1.32 million comparable blocks in the three sample markets; the exceptions are 78 blocks on one coin inside the first thirty-one minutes, worth at most 0.8 basis points of the mid. The gate opened 115 seconds after the empty-book restart, hours before the production window began, so the restart cost no production time.

The downstream screens differ by exercise. Markout construction requires fresh pre-event and endpoint states and sets gate-suppressed endpoints to missing. The forecasting grid truncates targets at UTC-day boundaries but carries the last emitted block state forward and does not apply the markout-specific staleness screens. This makes collection continuity especially important for the forecasting sample; Table A.1 exposes every gap in that window.

A.3 Audit

The audit comprises three components. *Fill pairing:* of 27,902,380 crossed (taker) fill records on July 1–27, every one pairs with a maker-side record at the same (day, market, trade identifier)—zero unmatched legs, by count and by notional. *Archive integrity:* a SHA-256 manifest covers every archived hourly shard of all six streams—the five node streams and the oracle sample (180 day-stream shard sets). *Collection continuity:* a forecasting result could be distorted if the archive selectively missed active periods. The day-by-day block census in Table A.1 weighs against this concern. Twenty-two of the thirty collection days show a

maximum intra-day block gap of three or fewer with no gap of ten blocks or more. Every fitting-window day (July 11–20) satisfies that block-gap criterion, as does the evaluation window apart from a sixteen-block gap on July 25 and a 28,326-block gap on July 27. The remaining interruptions are documented: larger block gaps in the scoring window (July 1–2), the truncated shards described above, and the July 28–29 node outage, which appears as a 1,119,241-block discontinuity across that boundary. July 30 is continuous once the second node's hours are merged, so the 20:00 UTC splice introduces no break in the block sequence.

Table A.1: Collection audit by day: block-level continuity

Day	Hours	Blocks (m)	Max gap	≥10	Absent (%)	Day gap	Trunc.
Scoring window: July 1–10							
7/1	24	1.150	27,900	1	8.3	—	0
7/2	24	1.097	59,415	3	12.8	-1	1
7/3	24	1.184	4	0	5.8	-1	2
7/4	24	1.151	7	0	5.1	0	5
7/5	24	1.128	3	0	5.3	0	3
7/6	24	1.132	4	0	5.7	-1	3
7/7	24	1.138	3	0	5.8	0	1
7/8	24	1.147	3	0	5.5	0	1
7/9	24	1.158	3	0	5.2	1	1
7/10	24	1.154	3	0	5.2	1	1
Fitting window: July 11–20							
7/11	24	1.138	3	0	5.1	-1	0
7/12	24	1.148	3	0	5.2	0	0
7/13	24	1.148	3	0	5.8	0	0
7/14	24	1.147	3	0	5.6	0	0
7/15	24	1.142	3	0	5.6	0	1
7/16	24	1.146	3	0	5.4	0	2
7/17	24	1.141	3	0	5.7	0	0
7/18	24	1.147	3	0	5.3	-1	0
7/19	24	1.151	2	0	5.2	0	0
7/20	24	1.149	3	0	5.3	0	3
Evaluation window: July 21–27							
7/21	24	1.149	3	0	5.2	0	4
7/22	24	1.148	2	0	5.0	-1	0
7/23	24	1.147	3	0	5.0	0	0
7/24	24	1.151	3	0	4.9	0	0
7/25	24	1.152	16	1	4.6	1	0
7/26	24	1.152	3	0	4.8	0	0
7/27	24	1.100	28,326	2	9.4	1	0
Post-sample: July 28–30							
7/28	7	0.293	2	0	4.9	-1	0
7/29	20	0.871	22,008	2	8.5	1,119,241	0
7/30	21+10	1.177	3	0	4.8	0	0

Notes: Census of distinct block numbers in the archived order-status and book-diff envelopes of both collection nodes, grouped by the window each day belongs to. July 28 is in the post-sample block but is excluded from the sample by the collection outage. Hours counts hourly shards carrying at least one block, primary node plus (after +) the second node. Max gap is the largest jump between adjacent observed block numbers within the day; ≥10 counts jumps of at least ten blocks. Absent is the share of block heights missing from the day's observed-height span. Day gap reports missing heights between adjacent daily spans; −1 denotes a duplicated boundary height and 0 denotes contiguity. Trunc. counts source shards decoded only through their verified last good record.

All fitting-window days have maximum gaps of three blocks or fewer; the evaluation-window gap exceptions are July 25 (16 blocks) and July 27 (28,326 blocks). July 30 is continuous after merging the second node from 20:00 UTC.

B Additional Results and Robustness

B.1 Peer-Adjusted Wallet Score

The main text defines wallet toxicity by the raw ten-second markout. This subsection repeats the toxicity validation and forecasting race with a peer-adjusted score, which measures how much more a wallet's orders mark out than the orders of *other* wallets trading the same coin in the same minute. The peer benchmark removes anything common to the market at that instant, so the comparison answers the natural objection that the identity block is picking up moments when everyone does well rather than wallets who do well.

The ranking survives peer adjustment, and persists better. If the raw ranking merely captures market-wide timing, removing the same-coin, same-minute component should weaken its persistence. Instead, the peer-adjusted rank has a Spearman ρ of 0.62 across adjacent ten-day windows, above the raw score's 0.52 (Figure B.1(a)).

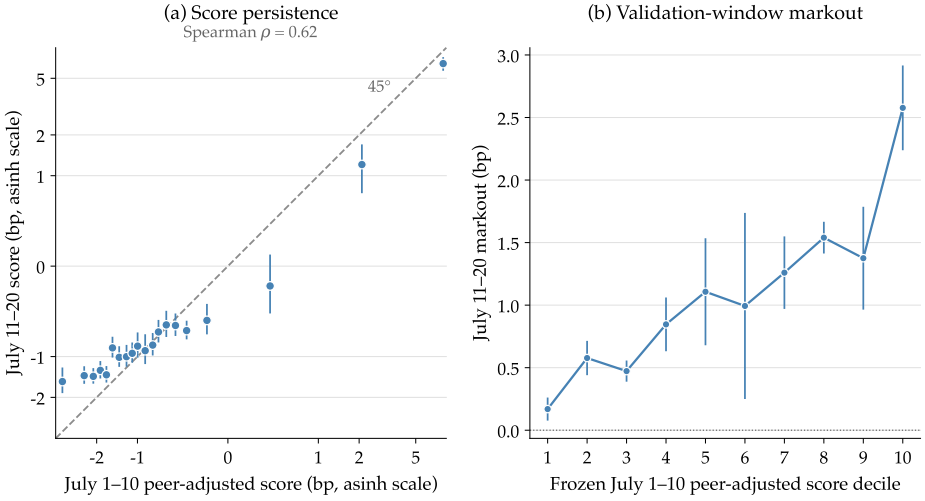


Figure B.1: Wallet toxicity using the peer-adjusted score. Panel (a) shows score persistence; Panel (b) shows validation-window markouts by frozen-score decile. Bars are 95% confidence intervals.

The peer-adjusted score also retains the economically relevant upper tail: validation markouts rise sharply across the top deciles (Figure B.1(b)). The accompanying decile portrait in Table B.1 further shows that the top decile pays the lowest median taker fee.

Table B.1: Wallet deciles under the peer-adjusted score

	Scoring window (July 1–10)				Validation window (July 11–20)
Decile	Wallets	Orders	Median order ($)	Median fee (bps)	Markout (bps)
D1	206	56,546	1,852	4.50	0.17
D2	206	77,396	1,747	4.50	0.58
D3	206	62,668	1,078	4.50	0.47
D4	207	118,149	578	4.50	0.85
D5	206	44,405	177	4.50	1.11
D6	206	55,167	100	4.50	0.99
D7	207	115,782	82	4.50	1.26
D8	206	178,964	58	4.50	1.54
D9	206	67,004	50	4.50	1.38
D10	206	242,721	1,007	2.97	2.58

Notes: Deciles use the July 1–10 peer-adjusted score and are never re-sorted; the peer benchmark is the notional-weighted mean markout of other wallets' orders in the same coin and minute, so the score measures excess markout rather than level.
Wallets, orders, median order, and median fee are measured in the scoring window; markout is the notional-weighted ten-second markout measured out of sample during July 11–20, which is the only statistic the peer-adjusted arm carries for the validation group. D10, the toxic decile, is set off by a rule.

The predictive gain survives, at roughly half the size. If common market timing drives the forecasting result, peer adjustment should also eliminate the identity gain. Against the identical public benchmark, Table B.2 shows that the short-horizon gain remains positive and precisely estimated: at one second it is 6.8% for the top decile, roughly half the raw score's 13.2%. Because peer adjustment removes the same-coin, same-minute component by construction, the remaining gain shows that the identity block is not solely a proxy for common market timing.

Table B.2: Return prediction under the peer-adjusted score

Horizon	Ridge (benchmark) + Identity R^2 (%)	Ridge (benchmark) Gain (%)	Ridge (benchmark) t	Gradient-boosted trees + Identity R^2 (%)	Gradient-boosted trees Gain (%)	Gradient-boosted trees t
Panel: Frozen score, top decile						
0.2s	7.46	+7.9	5.6	20.69	+7.4	5.5
0.5s	10.86	+6.9	5.0	22.28	+6.1	6.6
1s	11.62	+6.8	5.1	20.27	+4.1	5.6
2s	11.23	+5.9	4.2	16.91	+3.4	6.2
5s	9.62	+4.9	4.2	12.33	+2.8	5.7
10s	7.49	+4.3	4.4	8.72	+2.0	5.0
30s	3.67	+3.3	4.1	3.84	+2.7	3.9
Panel: Frozen score, top quintile						
0.2s	7.57	+9.6	5.5	20.94	+8.7	5.3
0.5s	11.04	+8.7	5.3	22.48	+7.1	5.5
1s	11.78	+8.3	5.4	20.35	+4.5	4.8
2s	11.34	+7.0	4.6	16.92	+3.5	5.2
5s	9.68	+5.5	4.6	12.31	+2.6	5.8
10s	7.53	+4.8	4.7	8.77	+2.6	4.9
30s	3.69	+3.8	4.4	3.83	+2.4	2.1
Panel: Rolling ten-day score, top decile						
0.2s	7.65	+10.7	4.2	20.05	+4.1	1.8
0.5s	11.05	+8.8	3.8	21.54	+2.6	0.8
1s	11.74	+7.9	4.3	20.02	+2.8	1.8
2s	11.25	+6.2	4.1	16.83	+3.0	4.2
5s	9.59	+4.6	4.8	12.27	+2.3	3.8
10s	7.46	+3.9	4.5	8.71	+1.9	3.1
30s	3.66	+2.8	3.1	3.86	+3.2	3.1
Panel: Rolling ten-day score, top quintile						
0.2s	7.87	+13.9	5.5	21.20	+10.1	3.4
0.5s	11.36	+11.8	5.2	22.72	+8.2	4.7
1s	11.98	+10.1	5.4	20.52	+5.3	5.1
2s	11.41	+7.6	4.8	17.02	+4.1	5.5
5s	9.65	+5.2	4.7	12.37	+3.1	4.8
10s	7.49	+4.3	4.6	8.73	+2.2	4.1
30s	3.66	+2.9	3.3	3.81	+1.8	2.1

Notes: Table 4 re-run with the wallet cohort defined by the peer-adjusted score instead of the raw score, over the full grid of score definitions and tail cuts. Scores are estimated July 1–10, models fit July 11–20, and evaluation is July 21–27. The *+ Identity* columns report out-of-sample R^2, Gain $= R^2_{+\text{Identity}}/R^2_{\text{Anonymous}} - 1$ is the relative increase, and t uses seven paired daily MSE differences.
The anonymous benchmark does not depend on how the cohort is selected, so it is the one printed in Table 4 and is not repeated here.

B.2 Cross-Venue Latency

Table B.3 reports the diagnostic summarised in Section 3.2, decile by decile: how often each decile's orders arrive on the side of the oracle–midpoint gap, the mean signed gap they trade

Table B.3: Cross-venue latency diagnostic by wallet decile

	All orders			Gap $\leq$ 1 bp	
Decile	Orders	Aligned (%)	Mean gap (bp)	Orders	Markout (bp)
D1	36,399	57.9	-0.62	6,815	+0.16
D2	57,184	56.0	-0.26	13,344	+0.10
D3	79,017	40.7	-0.64	16,959	-0.19
D4	61,822	45.4	-0.38	16,330	-0.03
D5	45,888	50.9	-0.86	10,000	+0.11
D6	85,933	45.0	-0.76	24,129	+0.11
D7	40,285	44.1	-1.30	7,861	+0.11
D8	36,758	46.8	-1.72	8,172	+0.63
D9	180,508	53.2	-0.77	40,922	+0.95
D10	408,053	41.3	-1.45	78,918	+2.12

Notes: The venue's oracle price aggregates external venues and is sampled every ten seconds; the gap is that oracle price less the local midpoint immediately before each aggressive order. *Aligned* counts orders arriving on the gap's side, which is what latency arbitrage would produce: a wallet reacting to an external move buys when the oracle sits above the local mid. *Mean gap traded into* is the signed gap in basis points, negative when the order trades against the displacement. The final two columns restrict to orders facing a gap of at most one basis point, where there is nothing to arbitrage; these are 22% of all orders. Fama–MacBeth daily D10−D1 markout coefficients over July 11–20 average 1.72 without gap controls and 1.75 with them. Because the oracle can be up to ten seconds stale, the test bounds a slow cross-venue channel but cannot rule out a subsecond one.

into, and the markout that survives in the subsample where the gap is at most one basis point.

B.3 Tail Cut and Score Definition

The forecasting result would be fragile if it depended on the exact tail cut or on freezing the score after July 10. Table B.4 replaces the top decile with the top quintile and the frozen score with a rolling ten-day score. Neither design choice materially changes the gain at any reported horizon.

Table B.4: Identity gain by tail cut and score definition

Horizon	Ridge (benchmark)			Gradient-boosted trees		
	+ Identity R^2 (%)	Gain (%)	t	+ Identity R^2 (%)	Gain (%)	t
Panel: Frozen score, top quintile						
0.2s	8.15	+17.9	7.6	21.57	+12.0	5.6
0.5s	11.73	+15.5	7.7	23.10	+10.0	5.2
1s	12.30	+13.1	8.4	20.78	+6.7	5.2
2s	11.68	+10.2	8.8	17.09	+4.6	5.7
5s	9.82	+7.1	8.4	12.40	+3.4	5.0
10s	7.57	+5.5	11.2	8.79	+2.8	4.8
30s	3.67	+3.2	5.1	3.92	+4.7	4.4
Panel: Rolling ten-day score, top decile						
0.2s	8.13	+17.7	6.0	20.94	+8.7	3.9
0.5s	11.69	+15.1	5.9	22.71	+8.1	4.1
1s	12.26	+12.7	6.5	20.65	+6.0	5.0
2s	11.61	+9.6	6.0	17.11	+4.7	5.9
5s	9.76	+6.3	6.2	12.37	+3.2	5.5
10s	7.52	+4.8	6.9	8.77	+2.6	5.2
30s	3.64	+2.3	6.0	3.88	+3.6	3.9
Panel: Rolling ten-day score, top quintile						
0.2s	8.07	+16.7	5.9	21.36	+10.9	5.6
0.5s	11.62	+14.4	5.8	22.97	+9.4	5.2
1s	12.20	+12.2	6.3	20.70	+6.3	5.4
2s	11.59	+9.4	6.5	17.12	+4.7	5.5
5s	9.76	+6.4	6.4	12.40	+3.3	5.1
10s	7.54	+4.9	7.9	8.79	+2.9	5.0
30s	3.65	+2.6	4.3	3.86	+3.2	2.9

Notes: Alternatives to the headline specification of Table 4, which reports the frozen score at the top decile. Sample, targets, variable blocks, learners and t construction are identical, so the anonymous benchmark is common to every panel and is the one printed in Table 4 rather than repeated here. The *+ Identity* columns report out-of-sample R^2 and Gain is the relative increase over that common benchmark.

B.4 Matched-Wallet Placebo

If size and activity alone explain the July result, a matched cohort should reproduce the toxic cohort's forecasting gain. No matched cohort reproduces the gain through ten seconds. At one second, the real top-decile R^2 increment is 1.44 percentage points against 0.88 for the best of 200 activity-matched draws, so the best matched cohort reaches 61% of the real increment. The rank p is 1/201 at every horizon through ten seconds (Table B.5). At thirty seconds the decile's increment of 0.11 percentage points falls inside a placebo range reaching 0.14 ($p = 0.16$), while the top quintile still exceeds every draw; the quintile is the wider cohort and averages over more wallets, which is what keeps its increment above the ceiling once the signal is small. Exact-cell donor limits make each draw smaller than the full toxic cohort, so

the rank is descriptive rather than an equal-size randomization test.

Table B.5: Matched-wallet placebo: the toxicity ranking against 200 activity-matched cohorts

		Real	200 matched cohorts, ΔR^2 (pp)				
Horizon	Anonymous R^2 (%)	ΔR^2 (pp)	mean	p95	max	max/real	p
Top decile							
0.2s	6.91	+1.291	+0.389	+0.489	+0.579	45%	0.0050
0.5s	10.16	+1.601	+0.596	+0.741	+0.851	53%	0.0050
1s	10.88	+1.435	+0.642	+0.773	+0.877	61%	0.0050
2s	10.60	+1.062	+0.534	+0.651	+0.690	65%	0.0050
5s	9.17	+0.617	+0.355	+0.443	+0.471	76%	0.0050
10s	7.18	+0.374	+0.229	+0.296	+0.308	82%	0.0050
30s	3.56	+0.107	+0.083	+0.124	+0.136	127%	0.1642
Top quintile							
0.2s	6.91	+1.239	+0.199	+0.301	+0.313	25%	0.0050
0.5s	10.16	+1.574	+0.289	+0.435	+0.451	29%	0.0050
1s	10.88	+1.426	+0.295	+0.383	+0.398	28%	0.0050
2s	10.60	+1.078	+0.215	+0.305	+0.317	29%	0.0050
5s	9.17	+0.647	+0.157	+0.221	+0.229	35%	0.0050
10s	7.18	+0.392	+0.106	+0.147	+0.157	40%	0.0050
30s	3.56	+0.113	+0.030	+0.053	+0.061	54%	0.0050

Notes: Panels give the identity block built from the top decile and the top quintile of the frozen toxicity ranking. Cohort quote features incorporate every block-close update before being carried to the 100ms grid, matching the headline feature construction.

Draws match the real cohort cell by cell on deciles of scoring-window notional crossed with order count, sampled without replacement; cells where the cohort exhausts the donor pool are filled as far as they go, so matched cohorts are slightly less extreme on activity than the real one.

Ridge benchmark, fit July 11–20 with expanding-day validation, evaluated July 21–27; the placebo mean, p95 and max summarize the same increment across the 200 draws. $p = (1 + \#\{\text{placebo} \geq \text{real}\})/201$, so $p = 0.0050$ is the floor attained when no draw reaches the real increment, and the max/real column reports the largest draw as a share of that increment. Because some cells exhaust their donor pool, the rank describes these generated cohorts rather than an exact equal-size randomization test.

C Replication on December 2025

The core design is repeated on an independent public dataset: the December 2025 release of Hyperliquid BTC, ETH, and SOL, scored December 1–10, fit December 11–20, and evaluated December 21–31. The score definitions, predictor blocks, learners, and horizons are unchanged. We report the persistence, forecasting, and matched-cohort exercises; the order-arrival timing test is specific to July. A useful replication should expose the design to different market conditions rather than simply extend the original sample. The December sample provides this variation. Relative to July, Table C.1 documents a deeper record of a narrower market set: three markets over a full month carry more messages and more taker notional

than July's ten over 27 days, on fewer wallets and fewer aggressive orders. Spreads are wider throughout, so the replication is not a re-run under easier conditions.

Table C.1: Level-4 sample by perpetual market, December 1–31, 2025

Market	L4 messages (bn)	BBO updates (m)	Agg. orders (m)	Taker fills (m)	Wallets (thousands)	Notional ($bn)	Spread (bps)
BTC	15.08	1.91	4.28	11.92	64.7	75.1	0.38
ETH	7.82	1.83	3.62	6.84	40.2	42.6	0.84
SOL	3.35	1.58	1.62	3.10	34.8	9.8	1.08
All	26.25	5.32	9.53	21.86	89.8	127.5	0.68

Notes: L4 messages combine raw order-status and book-diff records; BBO updates are gate-valid blocks in which the best quote changed. Aggressive orders, wallets, notional and spread come from the aggressive-event tape and taker fills are taker legs; spread is the mean quoted bid–ask spread immediately before aggressive orders. Wallets are distinct within market and deduplicated across markets in the All row.
The sample covers the three markets of the public December 2025 release, which carries accepted and rejected order statuses in separate archives; both are counted, matching the July node stream.

The score is persistent and the tail is stable. If the July ranking captures a stable wallet attribute, rank persistence should survive in a different quarter and reconstruction. Across adjacent December windows, Spearman ρ is 0.47, close to July's 0.52 (Figure C.1(a)).

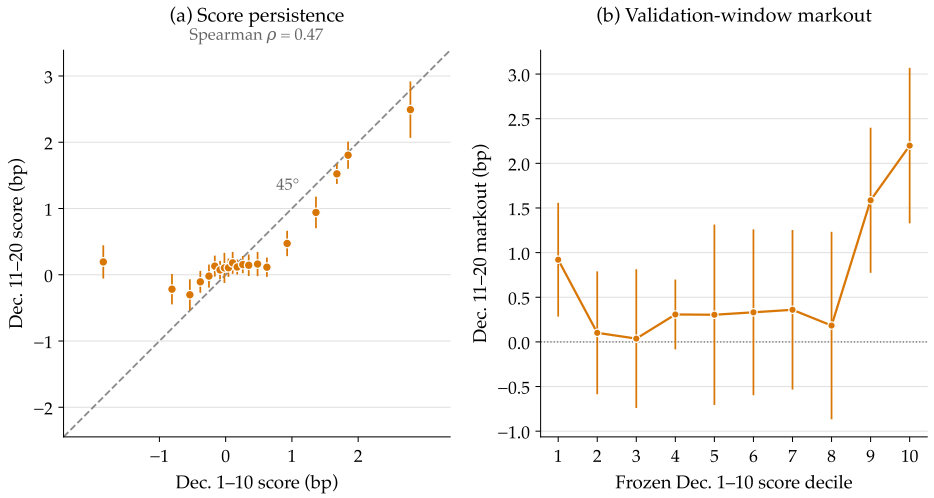


Figure C.1: December 2025 replication of wallet-score persistence and adjusted validation-window markouts. Bars are 95% confidence intervals.

Rank persistence matters because it again translates into an economically distinct upper tail. The scoring-window decile ladder is monotone, and validation markouts rise sharply at the top (Figure C.1(b)). The top decile's ten-second markout remains stable, and it holds a steady quarter of all discretionary taker notional across the two windows, 26.7% and 26.0%, close to July's 31.0% and 25.1% (Table C.2).

Table C.2: December 2025 wallet deciles: scoring window and out-of-sample validation

	Scoring window (Dec. 1–10)					Validation window (Dec. 11–20)				
Decile	Wallets	Orders	Markout (bps)	Median order ($)	Notional share (%)	Wallets	Orders	Markout (bps)	Median order ($)	Notional share (%)
D1	252	70,235	−1.51	136	1.0	211	45,558	−0.07	106	0.8
D2	253	101,242	−0.46	199	1.4	208	78,485	0.21	285	1.5
D3	253	139,770	−0.23	45	1.3	206	86,758	−0.06	84	0.8
D4	252	158,969	−0.08	32	2.1	183	103,876	0.02	13	0.4
D5	253	331,341	0.02	147	1.9	192	133,651	0.10	145	0.9
D6	253	184,267	0.13	84	1.5	213	87,907	0.15	81	0.9
D7	252	111,494	0.30	207	4.0	208	70,910	0.98	131	2.7
D8	253	127,782	0.50	748	5.8	221	81,911	0.21	634	5.6
D9	253	187,731	1.05	989	10.4	213	151,204	0.97	1,044	9.6
D10	252	1,179,099	2.25	731	26.7	240	828,640	2.34	810	26.0

Notes: Deciles are formed once on the frozen Dec. 1–10 raw score and never re-sorted. Both blocks report the same five statistics for qualifying aggressive orders on BTC, ETH, and SOL (TWAP and liquidation orders excluded): distinct active wallets, order count, notional-weighted ten-second markout, median order notional, and the decile's share of all discretionary taker notional in that window. That share column is taken against the whole window's taker flow rather than the scored subtotal, so it does not sum to one hundred percent; the residual is flow from addresses without a qualifying scoring-window history.
The scoring window is Dec. 1–10 and the validation window Dec. 11–20; the Dec. 21–31 evaluation window is not used here.

Identity predicts returns, on the same shape. If the forecasting gain is not specific to July, identity should again improve the anonymous benchmark out of sample. Table C.3 confirms this prediction: the identity block raises out-of-sample R^2 by +14.6% at the one-second headline horizon ($t = 3.9$), with larger gains at 0.2 and 0.5 seconds. As in July the gain is largest at the shortest horizons; unlike July it does not decay monotonically, ticking up again at thirty seconds, where the anonymous R^2 is below 3% and a relative gain is not comparable to the short-horizon cells. The anonymous benchmark is comparable to July's (12.36% against 10.88% at one second), so the gain is not an artifact of a weaker baseline in the replication sample.

Table C.3: December 2025 return prediction, top decile

Horizon	Ridge (benchmark)				Gradient-boosted trees			
	Anonymous R^2 (%)	+ Identity R^2 (%)	Gain (%)	t	Anonymous R^2 (%)	+ Identity R^2 (%)	Gain (%)	t
Panel: Frozen score								
0.2s	8.70	10.75	+23.5	5.3	26.30	28.76	+9.3	5.3
0.5s	12.71	15.24	+19.9	4.9	24.34	28.91	+18.8	5.4
1s	12.36	14.16	+14.6	3.9	21.16	25.15	+18.8	5.1
2s	10.50	11.58	+10.3	3.8	13.84	15.10	+9.0	6.3
5s	8.13	8.66	+6.4	3.4	9.08	9.64	+6.3	7.4
10s	6.03	6.38	+5.8	4.0	6.18	6.50	+5.3	7.9
30s	2.66	2.86	+7.8	4.8	2.68	2.80	+4.4	5.0
Panel: Rolling ten-day score								
0.2s	8.70	10.85	+24.7	5.1	26.30	28.60	+8.7	6.2
0.5s	12.71	15.28	+20.3	4.8	24.34	28.67	+17.8	6.4
1s	12.36	14.13	+14.4	4.0	21.16	24.74	+16.9	5.7
2s	10.50	11.50	+9.6	3.8	13.84	14.52	+4.9	8.4
5s	8.13	8.57	+5.4	2.9	9.08	9.50	+4.7	8.6
10s	6.03	6.30	+4.6	3.2	6.18	6.23	+0.9	5.1
30s	2.66	2.79	+5.0	3.4	2.68	2.76	+3.2	6.2

Notes: The December 2025 counterpart of Table 4, on the top decile of the score named in each panel. Scores use December 1–10, model fitting December 11–20, and evaluation December 21–31; variable blocks, targets and R^2 definitions are unchanged.
Anonymous is the benchmark without identity features and *+ Identity* adds the identity block; both report out-of-sample R^2. Gain $= R^2_{\text{+Identity}}/R^2_{\text{Anonymous}} - 1$ is the relative increase, and t uses eleven paired daily MSE differences.

The matched-cohort comparison holds. A genuine toxicity ranking should also remain more informative than wallets matched on size and activity. Against 200 cohorts drawn on empirical-decile cells, the December toxic cohort exceeds every draw at every reported horizon and both tail cuts (rank $p = 0.005$ throughout; Table C.4). The best of 200 draws reaches 62 to 83% of the real increment across the eight reported cells, a thinner margin than July's 25 to 82% over the same eight cells (Table B.5), but no draw reaches the real cohort in any cell. Exact-cell donor limits make the draws smaller than the full toxic cohort, so this rank is descriptive rather than an equal-size randomization test.

Table C.4: December 2025 matched-cohort comparison

		Actual toxic cohort			200 placebo cohorts, ΔR^2 (pp)			
Horizon	Anonymous R^2 (%)	ΔR^2 (pp)	Gain (%)	t	mean	p95	max	p
Top decile								
0.2s	8.73	+1.045	+12.0	3.3	+0.414	+0.641	+0.792	0.0050
0.5s	12.78	+1.876	+14.7	3.8	+0.779	+1.192	+1.448	0.0050
1s	12.38	+1.720	+13.9	4.0	+0.731	+1.122	+1.364	0.0050
10s	5.92	+0.226	+3.8	2.3	+0.073	+0.144	+0.180	0.0050
Top quintile								
0.2s	8.73	+1.072	+12.3	3.9	+0.409	+0.615	+0.660	0.0050
0.5s	12.78	+1.851	+14.5	4.4	+0.789	+1.199	+1.281	0.0050
1s	12.38	+1.689	+13.7	4.6	+0.742	+1.134	+1.196	0.0050
10s	5.92	+0.214	+3.6	2.6	+0.090	+0.168	+0.179	0.0050

Notes: Panels give the identity block built from the top decile and the top quintile of the frozen Dec. 1–10 toxicity ranking, with draws matched on cells crossing empirical deciles of scoring-window notional and order count. Exact matching is limited by non-toxic donor supply, so each top-decile (top-quintile) draw contains 154 (388) wallets while the actual cohort retains all 253 (505); the rank is descriptive rather than an equal-size randomization test.

Ridge models are independently tuned and refit on the 100ms decision grid, with identity quote state sampled each second and carried forward, so the actual comparator is a grid-specific refit rather than the headline model of Table C.3.

Gain $= R^2_{+\text{Identity}}/R^2_{\text{Anonymous}} - 1$ is the relative increase, and the placebo columns summarize the same increment across draws. $p = (1 + \#\{\text{placebo} \geq \text{real}\})/201$, so $p = 0.0050$ is the floor attained when no draw reaches the actual cohort. Scores use Dec. 1–10, fitting Dec. 11–20, and evaluation Dec. 21–31; t uses eleven daily MSE differences.

The two samples agree on the shape of the gain. December is a different quarter, collected and reconstructed by another party, and the design is applied to it without retuning. The replication reproduces both results: identity raises out-of-sample R^2 at every reported horizon (Table C.3), and the toxic cohort exceeds every activity-matched draw (Table C.4). A common mechanism should also produce a similar horizon profile. Figure C.2 confirms this: in both samples, the ridge identity gain is largest at the shortest horizons and falls monotonically through ten seconds. This shared decay is difficult to reconcile with a sample-specific artifact.

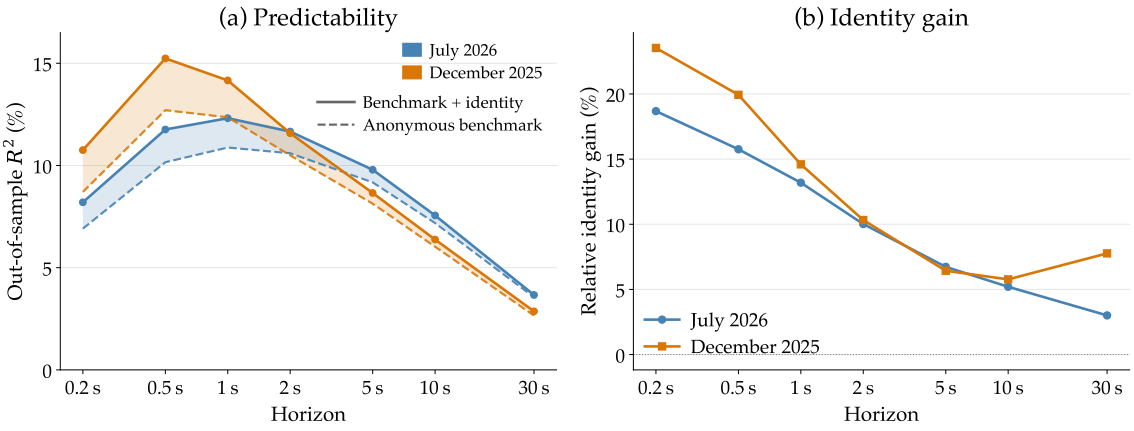


Figure C.2: Top-decile ridge results in July 2026 and December 2025. Panel (a) compares the anonymous benchmark with the model including identity; Panel (b) shows the relative identity gain.

D How Large Is the Increment?

A forecasting increment is easier to judge in payoff units than in R^2, so we convert the half-second increment into a midpoint payoff per unit of gross exposure. At each grid step the position is proportional to the anonymous or identity-augmented forecast. Activation thresholds are common ex ante budgets based either on trailing-ten-day or fitting-window forecast-magnitude quantiles; realized acted-grid shares are reported separately.

Per unit of gross exposure, the always-on identity edge exceeds the benchmark's by 7.0% ($t = 7.8$). That is close to the $\sqrt{11.76/10.16} - 1 = 7.6\%$ implied mechanically by the R^2 gain, which is the check the exercise has to pass: the conversion should reproduce the forecasting result in payoff units, and it does. Thresholded comparisons vary across activation budgets (Table D.1).

The exercise is frictionless and evaluated at the midpoint, with no book, queue, fill probability, spread, or inventory, so it is a unit conversion rather than evidence of capture. As a standalone taker strategy the gated rows' 0.25–0.76 basis points per unit of gross exposure would not cover the venue's 4.5-basis-point median taker fee. That comparison is the wrong benchmark for the mechanism the paper documents: the participant who can use the signal is a maker already quoting, for whom the relevant quantity is adverse selection avoided rather than a fee paid.

Table D.1: Frictionless conversion diagnostic by activation budget

	Payoff (bps)				
Budget	Anonymous	+ Identity	Increment	Relative gain	t
Panel A: rolling ten-day thresholds					
1%	0.754	0.752	-0.0012	-0.2%	-0.1
5%	0.470	0.495	+0.0253	+5.4%	2.9
10%	0.351	0.371	+0.0195	+5.5%	4.0
20%	0.249	0.264	+0.0144	+5.8%	5.0
Panel B: thresholds frozen on the fitting window					
1%	0.757	0.755	-0.0019	-0.2%	-0.2
5%	0.485	0.524	+0.0388	+8.0%	3.8
10%	0.361	0.385	+0.0232	+6.4%	5.3
20%	0.255	0.271	+0.0160	+6.3%	6.1
Panel C: no threshold (always active)					
Always active	0.131	0.140	+0.0091	+7.0%	7.8

Notes: Payoff is the realized half-second midpoint payoff per unit of gross exposure, $\sum \hat{y}\, y / \sum |\hat{y}|$ over acted moments (bps), July 21–27. Increment is the *+ Identity* minus *Anonymous* payoff difference in the same units, and Relative gain expresses it as a share of the *Anonymous* payoff.

Thresholds are per-coin $(1 - b)$-quantiles of $|\hat{y}|$ over the trailing ten days (Panel A) or the July 11–20 fitting window (Panel B); out-of-sample gates need not fire at the nominal budget, and the realized share of grid steps acted on is at or below it in every cell. t comes from the seven paired daily differences, consecutive decisions overlapping within the horizon.